

Kirill Mazur

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Education

Imperial College London

2021 - May 2026

Ph.D. student in Robotic Vision

- Dyson Robotics Lab; Supervisor: [Andrew J. Davison](#)
- Developing **Large-Scale Prior**-driven systems for real-time **SLAM** and **4D Reconstruction**
- Scene understanding through continual learning of a **neural field**-based map representation
- Published 4 papers at top-tier venues in Computer Vision & Robotics, including 3 as first author

Skolkovo Institute of Science and Technology, GPA: 3.83 out of 4.00

2018 - 2020

M.Sc. in Mathematics and Computer Science (with honors)

- Thesis: “Cloud Transformers”. Supervisors: [Victor Lempitsky](#) and [Gonzalo Ferrer](#)
- First year project: “Learnable point cloud descriptors for depth-only odometry and SLAM”

Yandex School of Data Analysis (YSDA), GPA: 4.84 out of 5.00

2016 - 2018

Computer Science (Masters-level degree school organized by Yandex)

Higher School of Economics (HSE), GPA: 9.22 out of 10.0

2014 - 2018

B.Sc. in Mathematics

Experience

Meta

Aug. 2024 - Feb. 2025

AI Research Scientist Intern at [Project Aria](#)

- Developed a **ViT** model for dense metric geometry estimation on the Aria device. Designed a **custom positional encoding** for cross-attention mechanism to improve prediction accuracy
- Trained dense point-map estimators at scale using PyTorch DDP (**>100k scenes and >1M images**). Built the data preparation pipeline and led iterative model refinement

Samsung AI Center

Jun. 2019 - Jul. 2021

Research Engineer at Virtual Human Telepresence Lab

- Published 2 first (co-)author papers at a top computer vision venue (ICCV).
- “Cloud Transformers”: Research on neural point cloud processing: a novel architecture for **multi-task neural 3D point cloud processing**: generation, completion, classification, and segmentation.
- “Point-Based Clothing Modeling”: Research on neural 3D clothes modeling and its visual try-on.

Yandex

Jun. 2018 - Sep. 2018

Intern ML Engineer at Computer Vision Lab

- Developed a model for Object Localization which reduced the response time by 75% and doubled a target product metric
- Adapted Tensorflow’s object detection project for Yandex Infrastructure

Publications

4D Primitive-Mâché: Glueing Primitives for Persistent 4D Scene Reconstruction

[Kirill Mazur](#), [Marwan Taher](#), [Andrew J. Davison](#)

[[project page](#), [video](#)]

Primitive-based 4D replayable reconstructions from casual videos.

(CVPR) *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026

KV-Tracker: Real-Time Pose Tracking with Transformers

[Marwan Taher](#), [Ignacio Alzugaray](#), [Kirill Mazur](#), [Xin Kong](#), [Andrew J. Davison](#)

[[project page](#), [video](#)]

Real-time pose tracking from feed-forward reconstruction models.

(CVPR) *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026

SuperPrimitive: Scene Reconstruction at a Primitive Level

[Kirill Mazur](#), [Gwangbin Bae](#), [Andrew J. Davison](#)

[[project page](#), [video](#), [code](#)]

Introduces a new representation for monocular scene reconstruction based on image segments.

(CVPR) *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024

Honorable Mention at CVPR Best Demo Track

Feature-Realistic Neural Fusion for Real-Time, Open Set Scene Understanding

Kirill Mazur, Edgar Sucar, Andrew J. Davison [project page, video]
Presents a new real-time high-dimensional feature fusion technique for open set scene understanding.
(ICRA) IEEE International Conference on Robotics and Automation 2023

Point-Based Clothing Modeling

Ilya Zakharkin*, Kirill Mazur*, Artur Grigorev, Victor Lempitsky [project page, code]
Presents a new technique for a visual try-on and clothes re-targeting for complex garments.
(ICCV) IEEE/CVF International Conference on Computer Vision 2021

Cloud Transformers: A Universal Approach To Point Cloud Processing Tasks

Kirill Mazur, Victor Lempitsky [project page, code]
Presents a new layer for 3D point clouds processing that achieved SoTA results on four various tasks.
(ICCV) IEEE/CVF International Conference on Computer Vision 2021

Awards	
Honorable Mention at CVPR Best Demo Track	2024
UK Global Talent Visa	2023
Dyson PhD Scholarship	2021
Skoltech Master’s Scholarship (fully covered tuition fees + bursary)	2019

Skills	
Frameworks	PyTorch, OpenCV, Open3D, CUDA, C++
Training	Deep Learning, Vision Transformers, Large-Scale Training, Vision Foundation Models (VGGT, SAM, DINO)
3D Vision	SLAM, Neural Rendering, Second Order Optimisation, SfM

Invited Talks	
Technical University of Munich (TUM)	July. 2024
Week-long research visit to Daniel Cremers’ lab	
Massachusetts Institute of Technology (MIT)	Jun. 2024
“Always-On Single-Image Priors for Spatial AI”	
Rank Symposium: Neural Rendering in Computer Vision	Aug. 2022
“Feature-Realistic SLAM”	

Academic Service	
Reviewer	
ECCV 2026, WACV 2025, ICCV 2025, CVPR 2026-2024, ICRA 2024-2023	
Imperial College London	Spring 2023-2024, Spring 2026
TA at Robotics course	
Skolkovo Institute of Science and Technology	Spring 2020
TA at Deep Learning course	
Higher School of Economics / Yandex School of Data Analysis	Sep. 2018 - Dec. 2019
Research seminar curator	
Yandex School of Data Analysis (YSDA)	Sep. 2018 - Jan. 2019
TA at Algorithms and Data Structures course	